

Passive Distributed Acoustic Sensing for Borehole Seismic Velocity Profiling: A 3 km Geothermal Well Field Trial

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Abstract We demonstrate passive borehole velocity profiling using coherent DAS on a 3 km geothermal well. Deconvolution interferometry with phase-weighted stacking recovers interval velocities from ambient and ground micro-vibrations without active sources, achieving <80 m/s mean absolute error at 0.5–1 km depth against a calibration shot ground truth.

Introduction

φ -OTDR turns standard optical fiber into a dense distributed acoustic sensing (DAS) array that resolves micro-strain variations with meter-scale sampling over kilometer-scale reach [1]. This sensing paradigm has now been widely adopted beyond telecom infrastructure, including earthquake detection [2], traffic monitoring [3], and field-deployed infrastructure surveillance [4]. A high-value extension is borehole velocity profiling, where a permanently installed fiber can provide continuous depth coverage without downhole electronics and repeated tool deployment, enabling cost-effective, repeatable monitoring of subsurface velocity changes for reservoir characterization, geomechanical assessment, and well integrity surveillance.

However, existing DAS-based velocity sensing approaches have two major limitations. Firstly, most deployed sensing fibers are laid on or near the ground surface [5], where the recorded signals are dominated by surface-propagating noise rather than deep body waves, limiting velocity recovery to shallow depths with insufficient resolution for characterizing deep geological structures. Secondly, borehole-deployed DAS fibers can overcome this depth limitation by directly sampling the subsurface wavefield, yet prior borehole DAS studies have relied on active vibration sources such as controlled surface excitations or downhole calibration shots [6], which are expensive, logistically complex, and difficult to repeat on demand.

In this work, we propose a passive DAS scheme for borehole velocity profiling based on deconvolution interferometry with phase-weighted stacking, which is validated on a 3 km geothermal well. The proposed method recovers accurate interval velocity models without any dedicated underground vibration source, relying solely on ambient microseismic background and ground noise signals.

Against a calibration shot ground truth, we achieve mean absolute error (MAE) <80 m/s at 0.5–1 km depth, and reliable passive recovery to ~ 2.0 km with MAE <250 m/s, beyond which the passive vibration source limits accuracy. We further show that the conventional passive approach, ambient noise cross-correlation (ANCC), fails in this borehole geometry with MAE exceeding 1600 m/s, as it retrieves only surface-wave energy rather than body-wave velocities, underscoring the necessity of deconvolution-based processing for passive borehole velocity recovery.

Methods

Fig. 1(a) shows the geological cross-section and borehole fiber deployment at Newberry Well 55-29, where the DAS fiber extends from the surface to 2.74 km depth alongside a surface injection pump and a calibration shot point at ~ 3 km. The proposed procedure, sketched in Fig. 1(b), extracts a depth-resolved velocity profile from DAS recordings without any dedicated active source. Raw IQ data from the interrogator yield the spatial phase gradient $\nabla\phi(z, t) = \partial\phi/\partial z$, proportional to the axial strain along the fiber, which serves as the primary observable. A zero-phase 4th-order Butterworth band-pass filter spanning 5–80 Hz retains the frequency band where acoustic wave arrivals carry the strongest signal-to-noise ratio.

Ground-truth reference — velocity model by a calibration shot (active source). A single active-source calibration shot at ~ 3 km depth provides a high-SNR DAS record with clear P-wave arrivals, yielding the most accurate velocity estimates. Frequency-wavenumber (F-K) filtering with a fan-shaped passband within $\pm 45^\circ$ of vertical separates the upgoing body-wave wavefield from tube waves and surface-related noise. Layer-stripping semblance then extracts velocities, serving as the ground-truth reference for validating the passive

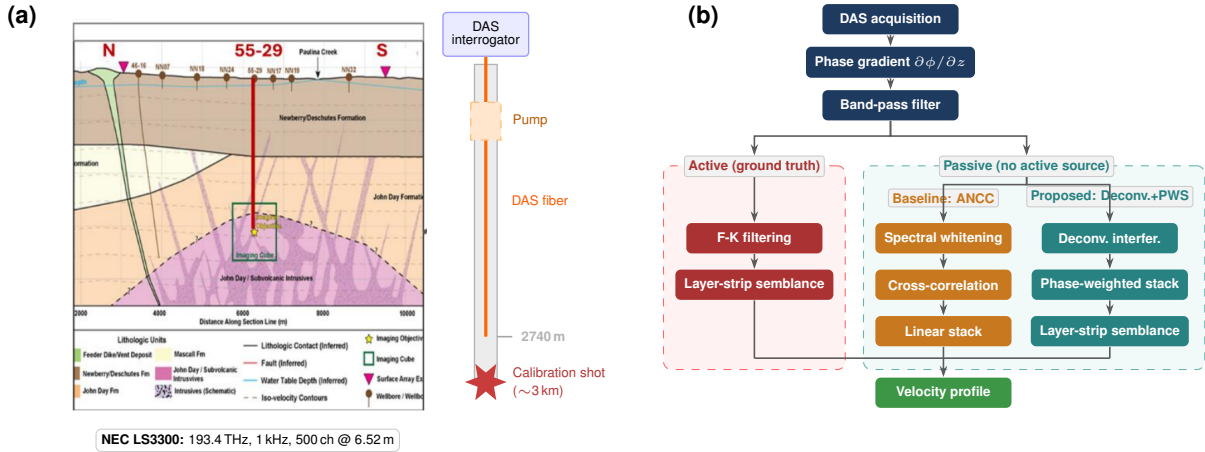


Fig. 1: (a) Geological cross-section of Newberry Well 55-29 [7] with borehole fiber deployment from surface to 2.74 km depth. (b) Signal processing procedure with two groups: the active-source calibration shot ground truth (left) and the passive methods requiring no active source (right), comprising the conventional ANCC baseline and our proposed deconvolution interferometry. All three methods extract a velocity profile.

methods. However, this approach requires deploying downhole explosives — a procedure that is expensive, logistically complex, and typically permitted only once during well completion, making it unsuitable for routine monitoring.

Conventional baseline — ambient noise cross-correlation (ANCC, passive). A natural alternative is to replace the active source with ambient noise. ANCC estimates inter-channel travel times by cross-correlating pairs of spectrally whitened DAS traces in the frequency domain:

$$C_{ij}(\tau) = \int \hat{u}_i(f) \hat{u}_j^*(f) e^{j2\pi f\tau} df \quad (1)$$

where $\hat{u}_i(f)$ and $\hat{u}_j(f)$ are the spectrally whitened DAS strain-rate signals at channels i and j , $\hat{u}_j^*(f)$ denotes the complex conjugate of $\hat{u}_j(f)$, and τ is the lag time. The peak of $C_{ij}(\tau)$ yields the travel-time difference between the two channels, from which the interval velocity is obtained. The resulting correlation gathers are linearly stacked over multiple reference channels to improve the signal-to-noise ratio. However, ANCC requires an isotropically diffuse noise field. When surface sources dominate, the retrieved correlations reflect shallow-propagating wave velocities rather than true body-wave velocities at depth.

Proposed method — deconvolution interferometry with phase-weighted stacking (passive). To overcome the limitation of ANCC while retaining fully passive operation, we replace cross-correlation with spectral-division deconvolution, which suppresses the shared source signature and isolates the inter-channel impulse response. For a reference channel at depth z_{ref} and a target channel at depth z , the deconvolved virtual-source gather is:

$$D(z, z_{\text{ref}}, f) = \frac{\hat{u}(z, f) \hat{u}^*(z_{\text{ref}}, f)}{|\hat{u}(z_{\text{ref}}, f)|^2 + \epsilon} \quad (2)$$

where $\hat{u}(z, f)$ and $\hat{u}(z_{\text{ref}}, f)$ are the Fourier spectra of the DAS traces at depths z and z_{ref} , and ϵ is a water-level regularization parameter set to 1% of mean spectral power that stabilizes the division at spectral nulls. The causal peak of the inverse Fourier transform of D gives the travel time between z_{ref} and z , effectively converting each reference channel into a virtual shot point. This formulation removes the common source wavelet, enabling velocity estimation even when the ambient noise field is non-diffuse.

Varying the reference channel and frequency band produces K independent gathers, combined via phase-weighted stacking (PWS):

$$S_{\text{PWS}}(t) = S_{\text{lin}}(t) \left| \frac{1}{K} \sum_{k=1}^K e^{j\varphi_k(t)} \right|^\nu \quad (3)$$

where $S_{\text{lin}}(t)$ is the linear stack, $\varphi_k(t)$ is the instantaneous phase of the k -th gather computed via the Hilbert transform, and $\nu = 2$ controls coherence weighting, suppressing incoherent noise while preserving coherent arrivals. Interval velocities are extracted via layer-stripping semblance over 50 m depth windows with a velocity search range of 1.5–6.0 km/s. Isotonic regression enforces a monotonically non-decreasing velocity profile, consistent with compaction-driven velocity increase with depth.

Experimental Setup

An NEC SpectralWave LS3300 φ -OTDR interrogator with sensing signal centered at 1550.12 nm is deployed at the wellhead of Newberry Volcano's ~3 km geothermal Well 55-29 [8], shown in Fig. 1(a). The DAS records IQ backscatter samples at a 1 kHz sampling rate along a 2.74 km standard single-mode fiber with 500 spatial channels at 6.52 m gauge length. The borehole penetrates a stratified volcanic sequence of Newberry tuffs, the John Day Formation, and deeper basaltic units [7],

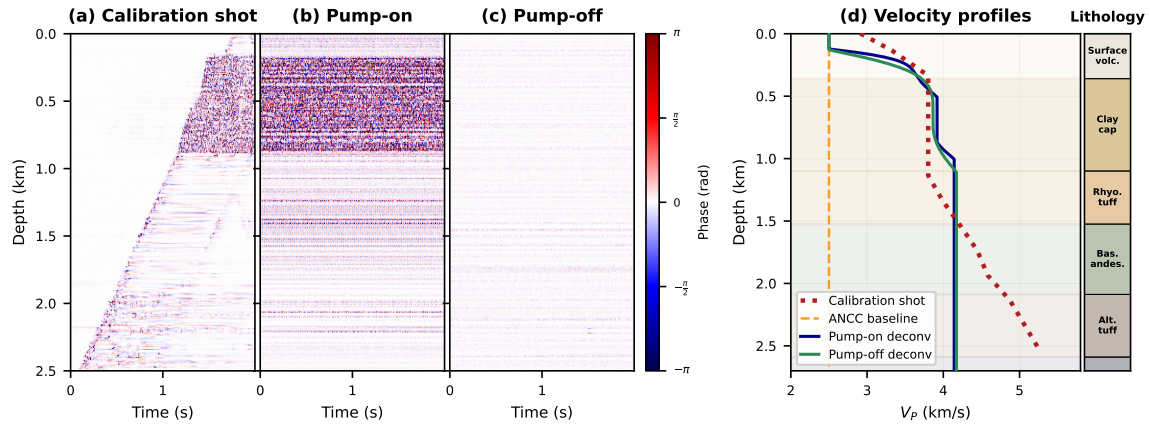


Fig. 2: (a) Calibration shot waterfall plot after F-K filtering. (b) Pump-on waterfall plot with sustained underground mechanical noise from the injection pump. (c) Pump-off waterfall plot with ambient microseismic signals. (d) Velocity profiles from all methods overlaid on known geological units [7] with lithology strip.

Tab. 1: MAE in m/s vs. calibration shot ground truth.

Depth	ANCC	Deconv pump-on	Deconv pump-off
0–0.5 km	994	285	356
0.5–1.0 km	1302	124	79
1.0–1.5 km	1403	236	241
1.5–2.0 km	1893	251	224
2.0–2.5 km	2477	834	804
Avg 0–2.0 km	1398	224	225
Avg 0–2.5 km	1647	360	352

providing a well-characterized multi-layer validation environment with documented velocities from 2.3 to 5.2 km/s.

Three source conditions are tested: (i) a calibration shot burned at ~ 3 km depth [8], providing high-amplitude vibrations as an active-source ground-truth test case; (ii) pump-on recordings over 3.75 hrs, where a surface injection pump coupling into fiber channels at ~ 220 – 860 m depth provided sustained broadband mechanical noise from 5 to 200 Hz; and (iii) pump-off recordings over 3.75 hrs, relying solely on ambient microseismic signals, which is the most challenging passive scenario. All data were segmented into 30-second windows with 50% overlap for stacking.

Results

After F-K filtering, the upcoming compressional and shear arrivals are clearly visible in the calibration shot waterfall plot in Fig. 2(a). The pump-on record shows sustained broadband noise from the injection pump, while the pump-off record contains only low-level ambient acoustic signals, shown in Fig. 2(b–c). Fig. 2(d) compares velocity profiles from all methods. The ANCC baseline produces a nearly flat ~ 2.5 km/s profile, dominated by surface-wave propagation. Both deconvolution results track the calibration shot closely at 0–1.5 km, rising from ~ 2.5 to ~ 4.2 km/s, corresponding to the transition from volcanic tuffs to denser basaltic formations at depth.

Tab. 1 quantifies the MAE against the calibration shot ground truth. ANCC yields MAE > 990 m/s

at all depths, with error increasing as the surface-wave bias grows. Both deconvolution conditions achieve ≤ 360 m/s overall MAE, with best performance of 124 and 79 m/s for pump-on and pump-off at 0.5–1.0 km. Notably, pump-off matches or exceeds pump-on at most depths, indicating that ambient microseismic energy suffices for passive velocity recovery when deconvolution properly removes the source signature. Furthermore, vibrations from the injection pump and associated subsurface infrastructure introduce additional noise into the velocity model, potentially degrading estimation accuracy for the pump-on condition.

Both passive profiles saturate at ~ 4.15 km/s below 1 km, correctly identifying the boundary between the Newberry Volcano tuffs and the underlying John Day Formation [7], while the calibration shot resolves all units to 2.7 km. Accuracy degrades below 2 km (MAE > 800 m/s) due to the finite virtual-source aperture, as coherent energy diminishes at greater depth separations [9].

Conclusions

We demonstrated passive borehole velocity profiling via coherent DAS on a 3 km geothermal well. Without any active source, deconvolution interferometry with phase-weighted stacking achieves MAE < 80 m/s at 0.5–1 km depth against a calibration shot ground truth, with reliable recovery up to ~ 2.0 km more broadly at MAE < 250 m/s over 0–2 km, while standard ANCC fails entirely with MAE exceeding 1600 m/s. These results establish that permanently installed borehole fiber can deliver continuous, source-free subsurface velocity monitoring to shallow-to-intermediate depths without dedicated downhole excitation.

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